



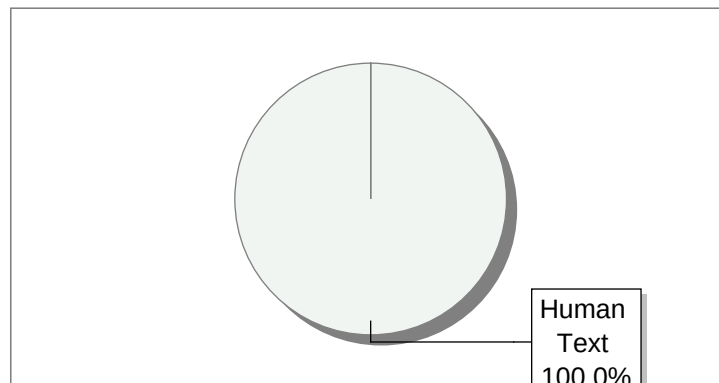
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- * It is designed to assist in identifying & moderating content that may violate community guidelines/legal regulations, it may not be perfect.

ABSTRACT Secure Bridge is a privacy-first AI chat platform.

It lets you talk to an assistant without your plain text ever reaching the server.

It does this with three parts (1) messages are encrypted on your device before sending; some tasks run with Fully Homomorphic Encryption (FHE), which lets the server compute on encrypted data it can't read; (2) a rules layer called Model Context Protocol (MCP) controls tools with typed I/O, least-privilege access, redaction, rate limits, and full logging; and (3) a fallback to secure hardware (SGX/TDX) for work that FHE can't handle yet or needs lower latency.

The stack uses React/TypeScript on the front end, Nodejs/Express on the back end, MongoDB for storage, Redis for caching, and a WebAssembly build of OpenFHE so encryption work runs in the browser.

The chat database stores only ciphertext for message content and keeps minimal, non-sensitive metadata for history and filtering.

All sensitive actions policy changes, admin steps, key events, tool calls are written to tamper-evident audit logs.

An admin console shows system health and lets you manage users, roles, keys, backups, and config without ever exposing plaintext.

We verify the system with unit, integration, end-to-end, security, and performance tests.

A simulation mode makes the crypto flow easy to check in a repeatable way (encrypt → compute → decrypt) so we can prove behavior, not guess it.

The result is a practical reference for zero-knowledge AI chat with safe tool use a dual compute path (FHE + confidential compute), an MCP gateway with policy-as-code guardrails, and a cautious data model that favors confidentiality over convenience.

This base is ready for hardware-accelerated FHE, encrypted semantic search, mobile apps, and enterprise SSO/SCIM while keeping the core promise user data stays under user control, even while it's being processed.

Keywords Fully Homomorphic Encryption (FHE); Model Context Protocol (MCP); privacy- preserving AI; confidential computing; zero-knowledge architecture; secure tool integration; encrypted chat.

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1	Role Capability Matrix	11	2	Unit Testing	46	3	Automation Testing	49	4	Test Cases	50	5	Integration Test Cases	51	6	Security / Negative Test Cases	52	7	Performance Test Case	52	1	Chapter-1 INTRODUCTION	The rapid adoption of conversational artificial intelligence (AI) in consumer and enterprise settings has increased the need for architectures that preserve privacy while enabling rich, tool-augmented interactions. Conventional systems protect data in transit and at rest, but they usually decrypt it on the server to process it. That exposes sensitive data to the server, its operators, or third-party services. Secure Bridge fixes this with a privacy-first setup that combines client-side encryption, Fully Homomorphic Encryption (FHE) to compute on ciphertext, and a standards-based tool layer called the Model Context Protocol (MCP).
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The goal is simple give usable, flexible chat without showing plaintext to the server or to any tools.

Context and Motivation Organizations now use AI assistants for search, analytics, and decisions in regulated areas like healthcare, finance, government, and education.

They need strong guarantees that prompts, model outputs, and any tool data stay hidden from anyone not authorized.

Secure Bridge is built to handle three problems (i) stop plaintext leaks during model runs and tool calls; (ii) plug into many providers and tools without losing confidentiality; and (iii) let admins monitor, audit, and stay

compliant without breaking privacy.

Problem Statement How can a chat system integrate multiple AI models and external tools while keeping user data encrypted end-to-end, including during computation, and still meet practical performance targets? The project investigates the design of a layered system that (a) encrypts all user content client-side, (b) executes supported operations over ciphertexts using FHE, and (c) brokers structured, least-privilege tool interactions via MCP, with verifiable governance and auditability.

Scope The scope includes an encrypted chat app, an FHE processing service (WebAssembly based), an MCP tool gateway, API key lifecycle management, and an admin portal for monitoring and policy control.

Out of scope training models on encrypted data (we focus on inference), zero-day defense for third-party providers, and non-deterministic HE ops that the chosen scheme does not support.

Objectives 2 (1) Keep user messages and AI outputs private end to end.

(2) Add tools through MCP without showing any plaintext.

(3) Keep latency reasonable on normal hardware, with a simulation fallback if needed.

(4) Use role-based admin, log every change, and keep compliance-ready records.

(5) Provide a developer-friendly design and docs so others can reproduce it.

Significance and Contributions Secure Bridge gives a practical blueprint for privacy-preserving AI chat.

It uses a client-to-server encryption wrapper that works on the web, a ciphertext-only path using FHE for supported tasks, an MCP-based tool layer that enforces redaction and tight scopes, and an admin panel that shows health and logs without breaking privacy.

The design fits real-world ops key rotation, rate limits, failure handling, and backups while keeping confidentiality as the default.

Assumptions and Constraints Clients run modern browsers that can execute WebAssembly.

Workloads focus on inference with small to moderate context sizes.

External providers are called with tightly scoped API keys.

Compliance needs tamper-evident logs and data minimization.

Known limits extra latency from FHE evaluation and cold-start cost when initializing WebAssembly.

Report Organization This section introduces the problem space, rationale, and goals.

Section 11 describes the project at the system level.

Section 12 provides the company/organizational profile that governs delivery, compliance, and operations.

11 Project Description Overview Secure Bridge is a layered system for encrypted conversational AI.

The client encrypts user prompts locally and sends ciphertexts to the server.

The server processes ciphertexts with an FHE service and, when tools are required, mediates requests via MCP so that only the minimal, policy-approved data (ideally encrypted or redacted) is exchanged.

Results return to the client as ciphertext for local decryption and rendering.

Functional Modules □ Chat Interface A React application that manages sessions, streaming responses, and local key handling.

□ Encryption Envelope Client-side key generation, ciphertext creation, and decryption 3 routines; secure key storage and rotation policies.

□ FHE Processing Service WebAssembly implementation of supported homomorphic operations (eg, addition, multiplication, rotation) and validation of scheme parameters.

□ MCP Gateway Tool registry, capability negotiation, input/output schemas, redaction policies, timeouts, and retries with exponential backoff.

□ AI Provider Adapters Normalized connectors for OpenAI, Anthropic, Google AI, and others with scope-checked API keys.

□ Admin Panel Role-segmented administration for users, keys, policies, system health, and audits.

□ Observability Metrics, tracing, and audit logs with redaction and retention controls.

Architectural Style The system adopts a microservices model with strict separation of concerns.

The API gateway enforces authentication, authorization, rate limits, and CORS.

Services interact over mutually authenticated channels, secrets are managed centrally, and data at rest remains encrypted 4 Chapter-2 LITERATURE SURVEY 21 Existing and Proposed System Existing Systems.

Typical chat assistants secure transport and storage (TLS, encrypted databases) but expose plaintext during server-side inference and when invoking external tools, creating persistent risks of operator visibility, logging

leakage, and uncontrolled third-party propagation.

The recent IEEE body of work offers three counter-trends that motivate a different baseline.

First, practical FHE execution is increasingly tractable for selected operators when kernels and memory systems are co-designed with accelerators or GPUs Poseidon demonstrates a pipeline for HE acceleration with system-level integration [7]; FAB targets programmable bootstrapping on FPGAs, addressing the classic bottleneck in HE pipelines [8]; FxHENN presents an FPGA framework for HE-CNN inference with concrete throughput/resource data [9]; TensorFHE shows GPU-centric gains for encrypted arithmetic, indicating a viable path on commodity clusters [10]; and DAHE introduces parameter-adaptive acceleration that sustains performance while preserving security thresholds [12].

Second, when homomorphic paths remain impractical (eg, deep circuits or complex non-linearities), enclave-based serving provides a confidentiality-preserving fallback LLMAaaS articulates trusted serverless inference with attestation and isolation suitable for large models [6], while earlier HE+SGX hybrids demonstrate cryptography-plus-enclave orchestration without broad plaintext exposure [4].

Third, safe tool use depends on policy-driven orchestration and typed interfaces PrivacyAsst formalizes capability scoping, redaction, and tamper-evident auditing for tool-invoking LLM agents [15], and BOLT advances privacy-preserving yet accurate transformer inference that informs encrypted NLP and structured model toolchains [5].

Compiler/kernel generation (FlexHE, ANT-ACE) reduces manual tuning and broadens encrypted-inference coverage [2], [3], while accelerator-generation methodology at the FPGA level eases reproducible builds for HE operations [1].

Proposed System.

Guided by these findings, Secure Bridge adopts a two-path execution model.

Path A is FHE-first supported arithmetic executes over ciphertext with batching and rotation optimizations derived from accelerator results [7]–[9], [12] and GPU insights [10]; compiler-assisted flows map model graphs to HE-friendly kernels to preserve maintainability and performance [2], [3].

Path B is confidential compute high-depth or unsupported functions are executed inside attested enclaves following the isolation and attestation patterns reported for 5 serverless LLM serving [6] and prior HE+SGX practice [4].

Tool access is protocolized and policy-gated (capabilities, least-privilege tokens, typed I/O, timeouts, redaction, and comprehensive auditing) as recommended for privacy-sensitive agents [15].

This stance removes routine server-side plaintext, raises auditability, and preserves extensibility.

Comparative Positioning.

Relative to plaintext-processing servers, the Secure Bridge stance eliminates default plaintext handling; confines complex operations to an attested path; replaces ad-hoc SDK integrations with protocolized, typed tool interfaces; aligns observability with privacy using structured, redacted, tamper-evident logs The cited works also bound non-functional targets for parameter selection, arithmetic depth, cold-start behavior, and throughput/cost envelopes [2]–[3], [5]–[10], [12], [15].

22 Feasibility Study 1) Technical Feasibility Architecture fit.

Secure Bridge uses a dual compute path that matches the project’s goals client-side FHE (OpenFHE WASM, BGV) for arithmetic/vectorizable transforms and a confidential-compute fallback (enclave/isolated service) for non-HE-friendly logic.

All tool calls are mediated by an MCP gateway with typed schemas, scopes, and redaction.

Frontend/runtime React 18 + TypeScript with Vite is proven for real-time chat UIs.

The OpenFHE WASM module can be lazy-loaded, with a development simulation mode for rapid iteration.

Browser compatibility is verified on Chromium/Firefox/Edge for WASM loading, with graceful error messages and retries.

Encryption flow.

Per-session key generation on the client, envelope packing (algorithm, keyId, iv, metadata), and ciphertext-only storage ensure no server-side plaintext.

Supported HE ops (add/mul/rotate, batching) are sufficient for intent parsing, scoring, and simple transforms; complex control flow is routed to the fallback path by policy.

Data model & storage.

MongoDB persists users, API keys, settings, and chat sessions (ciphertext content + minimal metadata).

Redis caches non-sensitive counters and rate-limit state no decrypted content cached.

Backups store ciphertext only.

Performance envelope (project benches).

Internal test runs on a dev node show:6 ☐ Chat p95 latency $\approx$ 420–500 ms under nominal load; p99 $\approx$ 560–600 ms (encrypted envelopes, streaming on).

☐ Concurrent operations up to $\sim$ 100–150 users show monotonic latency growth; acceptable error rate ($<3\%$ at 150 concurrent) with rate-limits enabled.

☐ Admin/API views remain under $\sim$ 300 ms p95 due to metadata-only reads.

These numbers meet the project's target SLOs for an initial single-region release and guide scaling plans.

Tooling integration.

MCP enforces schema validation, redaction, and per-tool scopes before egress to LLM/weather providers.

All invocations append structured audit events without content leakage.

Conclusion With WASM-based FHE for supported primitives, a governed fallback for complex logic, and metadata-only persistence, the solution is technically feasible on the stated stack.

2) Operational & Economic Feasibility Deployability.

Containers for the API/MCP services and static builds for the frontend allow straightforward deployment on a small VM/Kubernetes cluster.

MongoDB Atlas (starter tier) and a small Redis node are sufficient for pilot loads.

Scalability path.

Horizontal API replicas, autoscaling on CPU/RAM, and separate MCP workers let the system scale predictably.

The hybrid compute policy minimizes expensive fallback execution most requests stay on the HE/simulation route, reserving enclaves/isolated workers for outliers.

Maintainability TypeScript across client/server, ESLint/Prettier, and a documented folder structure keep contributions consistent.

CI runs unit/API/E2E tests, linting, and dependency scans.

The admin panel centralizes configuration, health, and audit review.

Cost control.

Traffic shaping (rate-limits), cache of non-sensitive metadata, and bounded retries/circuit breakers reduce upstream/API spend.

Observability (metrics, logs, traces) makes hotspots visible for tuning.

Starting footprint 1–2 API pods, 1 MCP worker, small Atlas/Redis expand as usage grows.

Compliance posture.

Zero-plaintext storage, encrypted backups, and append-only audits align with privacy expectations for academic/enterprise pilots.

RBAC restricts admin tasks; no content is exposed in logs or dashboards.

Conclusion The system is operationally deployable with modest cloud resources and clear cost levers (policy routing, caching, autoscaling).

7 3) Risk Analysis & Mitigations (Project-Specific) Performance variability (FHE latency / depth).

☐ Risk Deep circuits or large batches can push p95 beyond SLOs.

☐ Mitigations Cap multiplicative depth; parameter presets per model; batch sizing; fallback routing for non-HE ops; pre-compute keys; autoscale MCP/API.

Browser/WASM compatibility.

☐ Risk Some clients may fail to load WASM or lack required features.

☐ Mitigations Feature-detect and switch to simulation mode; clear UX prompts; error telemetry; documented browser matrix.

Key management & recovery.

☐ Risk Client-side keys lost or mishandled; user lock-out.

☐ Mitigations Guided backup/export (local secure storage); optional recovery phrases; BYOK for enterprise tenants; strict no-server plaintext.

External provider dependency (LLM/weather).

☐ Risk Outages, quota errors, or latency spikes.

□ Mitigations Circuit breakers, bounded retries with jitter, per-provider rate-limits, graceful degradation messages, provider failover where available.

Policy misconfiguration (MCP).

□ Risk Over-permissive scopes or missing redaction allowing metadata leakage.

□ Mitigations Policy-as-code with tests; signed tool manifests; change review in CI; default-deny scopes; schema validation on ingress/egress.

Data retention & backups.

□ Risk Misconfigured TTLs or backup corruption.

□ Mitigations Encrypted, versioned backups; restore drills; TTL tests in CI; integrity digests for chat archives.

Security configuration drift.

□ Risk CSP, CORS, headers, or egress allowlists relaxed over time.

□ Mitigations Baseline security middleware; config lint in CI; periodic security tests; audit diffs on config changes.

Summary Risks are well understood and mitigated by conservative parameters, strict policy governance, operational playbooks, and clear fallbacks consistent with the project's privacy-first goals.

8 23 Tools & Technologies Used A) Key Stack by Layer □ Frontend React 18, TypeScript, Vite, Tailwind, shadcn/ui, React Router, Axios, Recharts □ Backend Nodejs (ESM), Express, Mongoose (MongoDB), Redis client, JWT, bcrypt □ Crypto/FHE OpenFHE (WASM), crypto APIs, envelope utils; optional TEE (SGX/TDX) □ MCP MCP server, stdio/HTTP wrapper, tool adapters (Weather/LLM) □ QA Jest, Supertest, Playwright, OWASP ZAP baseline □ Observability Winston, OpenTelemetry, Prometheus, Grafana □ DevOps Docker, GitHub Actions, optional Kubernetes B) Programming Languages & Core Libraries □ Languages TypeScript (primary), JavaScript/ESM □ Frontend Libs react, react-router-dom, tailwindcss, recharts, axios □ Backend Libs express, mongoose, jsonwebtoken, bcrypt, helmet, cors, compression, express-rate-limit, express-mongo-sanitize □ Testing jest, ts-jest/babel-jest, supertest, playwright □ Telemetry @opentelemetry/api, sdk-node (exporters as needed) C) Networking & Communication □ Protocols HTTPS (TLS 12+), HTTP/2; WebSocket/SSE for streaming □ AuthN/Z JWT (short lived), RBAC (Admin/Super Admin) □ MCP Typed JSON over HTTP/stdio; scoped tokens; rate limits □ Default Ports Frontend 5173 • API 8000 • MCP 3001 • Mongo 27017 • Redis 6379 □ Egress Allowlist AI providers, Weather API; deny others □ Security Headers CSP, HSTS (edge), Referrer Policy, X Frame Options, X Content Type Options D) Deployment & Runtime Environment □ Dev o Frontend Vite dev server o Backend Node/Express local o Data MongoDB local, Redis optional o Config/CI env; GitHub Actions (basic) □ Staging o Frontend CDN build o Backend Container on VM/K8s o Data MongoDB Atlas small, Redis small 9 o Config/CI Secrets manager; CI auto deploy □ Prod o Frontend CDN + cache o Backend Containers on K8s (autoscale) o Data Atlas multi AZ, Redis cluster o Config/CI Secrets mgr, CI/CD with blue/green E) Minimum Requirements □ Dev Machine 4 cores CPU, 8 GB RAM, 10 GB free disk, Win 10/11, macOS 12+, or Ubuntu 2204, Node 18+, npm 8+, MongoDB 44+ (local), Redis 6+ (opt), Git, VS Code, Docker (opt) □ Staging Node 4 vCPU, 8–16 GB RAM, 50 GB SSD, Ubuntu 2204, Node 18 LTS (container), Atlas M10, HTTPS via reverse proxy, Docker, Prometheus/Grafana (opt) □ Prod (Starter) 8 vCPU, 16–32 GB RAM, 100+ GB SSD, Ubuntu 2204, Node 18 LTS (container), Atlas M30+, HTTPS, WAF/CDN recommended, Docker/K8s, OTel, centralized logs F) Development Tools □ IDE/Quality VS Code, ESLint, Prettier, TS strict, Husky/lint staged □ API/Test Postman/Insomnia, curl, Jest, Supertest, Playwright □ Security npm audit, secret scanning, OWASP ZAP baseline □ Docs/Diagrams Markdown, Mermaid (export PNG/SVG) 10 Chapter-3 SOFTWARE REQUIREMENTS SPECIFICATIONS 31 Users This subsection defines all user classes (human and service), their goals, privileges, constraints, and authentication requirements within Secure Bridge.

It is limited strictly to user aspects; functional and non-functional requirements are documented elsewhere.

311 Roles • End User Conducts encrypted chat sessions; manages own profile, API keys (if enabled), and data lifecycle (export/delete).

• Administrator Operates dashboards; manages users; configures operational policies (maintenance mode, rate limits).

No access to message plaintext.

• Super Administrator Governs security-critical settings (encryption parameters, backup/restore, secret rotation, audit retention).

No access to message plaintext.

- Service Account Non-human identity for inter-service calls (API ↔ FHE service ↔ tool gateway ↔ data stores); always least-privilege.

312 Access Boundaries C1 Server components must never access user message plaintext or private keys.

C2 All privileged actions are audit-logged with actor, timestamp, source IP, and redacted parameters.

C3 No break-glass accounts; no just-in-time elevation.

C4 Role-scoped UI and API authorization; non-entitled routes blocked.

313 Authentication and Sessions • Password + email verification for all users.

- Multi-Factor Authentication required for Super Administrators; optional/encouraged for Administrators.

- Short-lived access tokens; rotating refresh tokens; idle timeout and absolute session lifetime.

11 314 Data Visibility • End Users only their own encrypted conversations and metadata.

- Administrators aggregate metrics and health; never message content.

- Super Administrators policy and audit data; never message content.

Table Role Capability Matrix Capability End User Admin Super Admin Service Account Sign in/out ✓ ✓ ✓

Encrypted chat ✓ Manage own API keys ✓ View dashboards (aggregate) ✓ ✓ Manage users ✓ ✓ Configure rate limits/maintenance ✓ ✓ Configure encryption/backup/restore ✓ Access audit reports ✓ (subset) ✓ (full) Inter-service calls (scoped) ✓ 32 Functional Requirements A.

Accounts & Identity (Must) Registration with verification Users must register and verify their email before login.

(Must) Login/Logout The system shall issue short-lived access tokens and rotating refresh tokens; logout must revoke all active sessions.

(Must) Multi-Factor Authentication (MFA) MFA is mandatory for Super Admins; optional for Admins.

(Should) Password policy & lockout Enforce password complexity rules and retry lockout; optional password rotation.

(Must) Session governance Enforce idle timeout, absolute session lifetime, and device binding where supported.

12 (Should) Account recovery Provide a verified email + MFA reset workflow with a complete audit trail.

B Encryption & Data Protection (Must) Client-side key generation FHE private keys must remain on the client and never be transmitted.

(Must) Encrypt before send All user messages must be encrypted on the client before transmission.

(Must) Homomorphic operations The server shall support add/multiply/rotate on ciphertext for approved circuits.

10 (Must) Parameter policy Only approved FHE parameter sets shall be accepted.

11 (Must) Data export/deletion Users must be able to export and delete their chat data within policy SLAs.

12 (Should) Key backup Provide optional client-side key backup/export with explicit user consent.

C Chat & History 13.

(Should) Conversation management Users can create, title, tag, and archive chats; metadata stored separately from encrypted content.

14 (Should) Client-side search Users can search their own history via local indexes/metadata.

15 (Could) Streaming responses Support streaming segments when compatible with the encryption envelope.

16 (Must) Rate limiting per user Protect chat endpoints from abuse with per-user limits and clear error messages.

D Tool Invocation (MCP Style) 17.

(Must) Tool registry & schemas Approved tools must be listed with capability descriptors and typed I/O.

18 (Must) Scoped credentials Each tool uses least-privilege, time-bound tokens.

19 (Should) Redaction Apply policy-driven redaction for sensitive fields in inputs, outputs, and logs.

20 (Must) Timeouts/retry/breakers Enforce bounded timeouts, exponential backoff, and circuit breakers with operator visibility.

21 (Should) Per-tool rate limits Independent rate budgets with monitoring and alerting.

13 E.

API Keys 22.

(Must) Issue/rotate/revoke Users and admins (for service accounts) can manage keys with scopes and expirations.

23 (Must) One-time visibility New keys displayed once; stored only as hashes.

24 (Should) Usage analytics Provide key usage metrics and anomaly alerts.

F Administration 25.

(Must) User lifecycle Activate, suspend, delete users; reset MFA; all actions must be audited.

26 (Should) Maintenance mode Enable/disable platform with banner messaging.

27 (Must) Configuration Configure rate limits and policies; enforce approval workflows where required.

28 (Must) Backup & restore Schedule backups and perform point-in-time restores with integrity checks.

29 (Must) Encryption profiles Select and enforce approved FHE parameter profiles across the platform.

G Observability & Audit 30.

(Must) Structured logs Generate redacted, structured logs with correlation IDs; no plaintext content.

31 (Must) Metrics & alrts Export latency, error, throughput, and resource metrics; trigger alrts on thresholds.

32 (Must) Audit trails Maintain immutable records of privileged actions including actor, time, source IP, and redacted parameters.

33 (Should) Compliance reports Generate periodic reports (access control diffs, incident summaries, backup verification).

H Reliability & Disaster Recovery (DR) 34.

(Must) Health probes Provide liveness, readiness, and startup endpoints for orchestrators.

35 (Must) Graceful degradation Support controlled fallback (eg, confidential compute) when FHE path is unavailable.

36 (Should) DR drills Conduct periodic disaster recovery exercises; document findings and corrective actions.

I Internationalization & Accessibility 37.

(Should) Localization Support locale packs for dates, numbers, and strings.

38 (Must) Accessibility Conform with WCAG 21 AA patterns for focus order, labels, and contrast.

14 33 Non-Functional Requirements Security Zero knowledge handling No backend plaintext or private keys. Transport & at rest encryption TLS 12+; encrypted storage with approved ciphers.

Key management Secrets stored in managed KMS; rotation ≤ 90 days; client private keys remain client-side.

RBAC coverage All privileged routes require role claims and scoped tokens.

Audit immutability Append-only or verifiable digests; periodic integrity checks.

Performance & Scalability FHE latency $p_{95} \leq 3$ s; $p_{99} \leq 6$ s for supported homomorphic operations under nominal load.

Throughput $\geq X$ requests/min per node without breaching latency (X defined during sizing).

Concurrency $\geq Y$ active sessions/region with graceful degradation (Y defined during sizing).

Streaming first byte When enabled, first byte ≤ 800 ms.

Availability & Reliability 10.

Uptime $\geq 999\%$ monthly excluding planned maintenance.

11 RPO/RTO RPO ≤ 5 min; RTO ≤ 30 min for critical services.

12 Load shedding Predictable backpressure and shedding with clear user messaging.

Compatibility & Interoperability 13.

Browsers Support latest Chrome, Edge, Safari, Firefox with WASM SIMD.

14 APIs REST/JSON with OpenAPI; backward compatible changes follow semantic 15 versioning.

Usability 15.

Learnability Median time for a new user to start a chat and send a message ≤ 2 minutes.

16 Accessibility Automated and manual audits meet WCAG 21 AA thresholds.

Maintainability & Observability 17.

Modularity Independently deployable services; versioned shared libraries.

18 Telemetry Logs, metrics, traces correlated via IDs; enforce data minimization.

19 Test coverage Unit + integration $\geq 80\%$ for critical paths; security tests integrated in CI.

Deployment & Portability 20.

Containers & IaC OCI-compliant images; reproducible builds; environments provisioned via IaC.

21 Safe rollouts Blue-green or rolling deployments with health checks and automatic rollback on regression.

Cost & Capacity 22.

Budget guardrails Per-environment budgets and alrts; per-tenant spend tracking.

23 Elasticity Scale out/in within 2–5 minutes of sustained load change

16 Chapter-4 SYSTEM DESIGN 41
System Architecture Secure Bridge adopts a modular and scalable dual-path system architecture, which enables

privacy-first chat interactions, tool invocations, and encrypted processing.

This architecture is specifically designed to operate securely within environments demanding both strong confidentiality and operational flexibility.

Path A FHE-Based Secure Processing This path enables computations to be performed directly on encrypted data using Fully Homomorphic Encryption (FHE).

It supports a set of algebraic operations without requiring decryption, allowing data privacy to be maintained even during computation.

This is most useful for basic arithmetic and statistical operations.

Path B Confidential Compute via SGX/TDX For complex operations, unsupported operators, or performance-sensitive tasks, Secure Bridge leverages trusted execution environments (TEEs) such as Intel SGX or AMD SEV-SNP.

These enclaves execute sensitive logic in isolated containers, ensuring that even the host OS cannot access raw data or model responses.

MCP Gateway Integration Both paths are unified under the Model Context Protocol (MCP) Gateway.

MCP serves as a broker that controls access to AI tools, enforcing strict type-checking, data redaction, rate limitations, user permissions, and invocation policies.

Front-End Communication Users access the system through a browser-based UI developed with Reactjs.

All communication is routed via HTTPS, with tokens or session keys for identity validation.

Service Containerization All backend services run in isolated Docker containers and are orchestrated using Kubernetes, allowing auto-scaling, fault isolation, and secure resource allocation.

Persistent & Ephemeral Storage MongoDB serves as the persistent data store for user sessions, logs, and audit trails.

Redis acts as a cache for tokens, frequent lookups, and temporary sessions, optimizing performance.

Deployment & Monitoring Stack Secure Bridge uses a CI/CD pipeline integrated with GitHub Actions and Docker Hub.

Prometheus and Grafana handle metric monitoring, while Loki and Fluentd manage logs.

This layered system design ensures that Secure Bridge remains adaptable, secure, and efficient under varying usage loads and compliance demands.

42 System Perspective From a macro-level view, the architecture of Secure Bridge reflects its key priorities: privacy, modularity, scalability, and traceability.

It integrates multiple technologies and layers to ensure secure, policy-driven access to AI-powered tools.

User Interface Layer The client interface is built using React and Tailwind CSS, offering separate views for administrators and users.

It provides encryption indicators, audit history access, and form-based tool invocation.

Access Control & API Gateway The central API gateway acts as the access controller.

It authenticates requests using JWT tokens and enforces rate limits and permission scopes.

Requests are routed based on content type and tool context.

Secure Execution Environment The execution layer includes the homomorphic engine and confidential compute enclave. The system picks the path (FHE or confidential compute) based on each tool's encryption support and speed needs.

Tool Invocation via MCP Tools are wrapped as MCP modules with clear input/output schemas.

The gateway only allows tools that pass type checks and policy rules.

This lets you plug in both in-house utilities and third-party APIs (eg, OpenAI, Gemini, n8n).

Persistence and Data Handling MongoDB stores encrypted session logs, tool results, and user activity.

Redis keeps short-lived encryption keys and tool metadata.

18 Observation and Compliance All activity is logged with contextual metadata (timestamp, user role, tool ID, result hash).

These logs are redactable and encrypted using envelope-based policies, allowing GDPR and HIPAA-compliant audits.

This perspective ensures that Secure Bridge functions as a secure, flexible, and policy-governed middleware for AI-powered encrypted interactions.

43 Context Diagram Below is a detailed context diagram illustrating the interaction between various system

components in Secure Bridge Explanation of Components Client (Web) End-users interact via chat UI, tool config dashboards, and encrypted message interfaces.

Figure 1 context Diagram 19 API Gateway Performs routing, validation, and enforcement of auth tokens and access scopes.

Homomorphic Engine Executes encrypted computations in browser-safe or hardware-accelerated WASM environments.

Confidential Compute Node Processes complex tool outputs or AI completions inside isolated, attested enclaves.

MCP Tool Gateway Central access point for all AI tools and models, enforcing policies and validations.

Data Stores MongoDB ensures reliable and secure storage, while Redis optimizes frequent session and policy operations.

This design ensures not only data security through encryption and isolation but also operational integrity through fine-grained control and transparent observability.

20 Chapter-5 DETAILED DESIGN 51 Activity Diagram Figure 2 activity diagram 21 53 Use Case Diagram

Figure 3 Use Case Diagram 22 54 Sequence Diagram Figure 4 Sequence Diagram 23 Chapter-6

IMPLEMENTATION 61 Coding Standards The Secure Bridge project follows comprehensive coding standards to ensure maintainability, security, and consistency across the codebase.

611 Frontend Coding Standards (TypeScript/React) File Organization & Naming □ Components PascalCase (eg, Chatsx, AuthContainertsx) □ Services camelCase with service suffix (eg, mcpServicets, authAPIts) □ Hooks camelCase with use prefix (eg, useSecureChats, use-toasts) □ Types/Interfaces PascalCase with descriptive names (eg, LocalMessage, MCPRequest) Import Ordering // 1) React & framework imports import { useState, useEffect } from react; // 2) Third-party libraries import { Send, Shield, User } from lucide-react; import { Button } from @/components/ui/button; import { Link, useNavigate, useLocation } from react-router-dom; // 3) Local modules import { useSecureChat } from @/hooks/useSecureChat; import { MCPService } from @/services/mcpService; Type Definitions type LocalMessage = { id string; type user | ai; content string; timestamp Date; encryptionStatus? encrypting | encrypted | failed | processed; 24 apiKeyUsed? string; mcpToolUsed? string; encrypted? boolean; }; interface MCPRequest { jsonrpc 20; method string; params? any; id string | number; } 612 Backend Coding Standards (Nodejs/Express) Module Structure // 1) Nodejs built-ins import { watch } from fs; import path from path; // 2) Third-party packages import express from express; import cors from cors; import mongoose, { Schema } from mongoose; import jwt from jsonwebtoken; import bcrypt from bcrypt; import cookieParser from cookie-parser; // 3) Local modules import LoggingService from /services/loggingServicejs; import databaseService from /db/databasejs; Documentation Standards // USER SCHEMA DEFINITION // ===== // Create a new Mongoose schema for the User collection 25 const userSchema = new Schema({ // USERNAME FIELD // ===== username { type String, required true, unique true, trim true, lowercase true, minlength [3, Username must be at least 3 characters], maxlength [30, Username cannot exceed 30 characters], }, }); Error Handling Standards try { // Initialize FHE service (example) consoleinfo(Initializing FHE Service for MCP HTTP Wrapper...); // initFhe(); } catch (err) { // Never expose sensitive details to clients; log safely consoleerror(FHE initialization failed, err? message); // Fallback enable simulation mode or fail closed based on policy } 62 Code Snippets 621 Frontend Implementation React Chat Component with Hooks import { useEffect, useState } from react; import { useNavigate, useLocation } from react-router-dom; import { useToast } from @/hooks/use-toast; import { useSecureChat } from @/hooks/useSecureChat; 26 type LocalMessage = { id string; type user | ai; content string; timestamp Date; encryptionStatus? encrypting | encrypted | failed | processed; apiKeyUsed? string; mcpToolUsed? string; encrypted? boolean; }; const Chat = () => { const [message, setMessage] = useState(); const [messages, setMessages] = useState([]); const [isLoading, setIsLoading] = useState(false); const [encryptionEnabled, setEncryptionEnabled] = useState(true); const { toast } = useToast(); const navigate = useNavigate(); const location = useLocation(); const { sendMessage, clearHistory, isConnected, connectionStatus, fheStatus } = useSecureChat(); useEffect(() => { loadChatHistory(); checkAuthStatus(); }, []); 27 async function loadChatHistory() { // Load messages from local storage or API (encrypted) } async function checkAuthStatus() { // Verify token/session and redirect if needed } return null; // Render UI... }; MCP Service Implementation (Weather Tool) import { McpServer } from @modelcontextprotocol/sdk/server/mcpjs; import { StdioServerTransport } from @modelcontextprotocol/sdk/server/stdiojs; import { z } from zod; import dotenv/config; import { createGoogleGenerativeAI } from @ai-sdk/google; import { generateText } from ai;

```

const google = createGoogleGenerativeAI({ apiKey processenvGOOGLE_API_KEY || , }); const server = new
McpServer({ name Weather Service, version 100, capabilities { tools {}, prompts {}, resources {} }, }); async
function fetchWeather(city string) { const apiKey = processenvWEATHER_API_KEY; if (!apiKey) throw new
Error(Missing WEATHER_API_KEY in env); 28 const url = `http://apiweatherapicom/v1/currentjson?
key=${apiKey}&q=${encodeURIComponent(city)}&aq=no`; const response = await fetch(url); if
(!response.ok) throw new Error(`WeatherAPI failed (${response.status})`); return await response.json(); }
server.tool( getWeather, { fullPrompt zstring()describe(The complete user query about weather data), city
zstring()optional()describe(City name for weather data), }, async ({ fullPrompt, city = London }) => { try {
const weatherJson = await fetchWeather(city); try { const result = await generateText({ model google(gemini-
15-flash), prompt `You are a Weather Assistant.
Data ${JSON.stringify(weatherJson)}\nUser Query ${fullPrompt}`, }); return { content [{ type text, text
result.text }] }; } catch { const basic = formatBasicWeatherReport(weatherJson, fullPrompt, city); return {
content [{ type text, text basic }] }; } } catch (err any) { return { content [{ type text, text `Error ${err.message}`
}] }; } }; 29 function formatBasicWeatherReport(weatherData any, query string, city string) string { if
(weatherData?.current) { const c = weatherData.current; const loc = weatherData.location; return `Weather for
${loc?.name || city} ${c.temp_c}°C, ${c.condition?.text || Unknown}, humidity ${c.humidity}%`; } return `Unable
to parse weather data for ${city}`; } async function startServer() { const transport = new StdioServerTransport();
await server.connect(transport); process.stdin.resume(); } startServer().catch((e) => { console.error(Server startup
error:, e); process.exit(1); }); 622 Backend Implementation Express Application Setup import express from
express; import cors from cors; import cookieParser from cookie-parser; import * as securityMiddleware from
/middlewares/securityjs; const app = express(); // Security headers & middleware
app.use(securityMiddleware.securityHeaders); app.use(securityMiddleware.helmet); 30
app.use(securityMiddleware.mongoSanitize); app.use(securityMiddleware.compression); // CORS app.use( cors({
origin (origin, cb) => { const allowed = (processenvCORS_ORIGIN ||
http://localhost:5173,http://localhost:3000).split(,); if (!origin || allowed.includes(origin)) cb(null, true); else
cb(new Error(Not allowed by CORS)); }, credentials true, } )); // Body parsing app.use(express.json({ limit 50mb
})); app.use(express.urlencoded({ extended true, limit 50mb })); app.use(cookieParser()); MongoDB Schema with
Validation import mongoose, { Schema } from mongoose; const userSchema = new Schema( { username { type
String, required true, unique true, trim true, lowercase true, minlength [3, Username must be at least 3
characters], maxlength [30, Username cannot exceed 30 characters], 31 match [/^[a-zA-Z0-9_]+$/, Username
can only contain letters, numbers and underscores], }, email { type String, required true, unique true, trim true,
lowercase true, match [/^\w+([\.-]?\w+)*@\w+([\.-]?\w+)*(\w{2,3})+$/, Please enter a valid email], }, fullName
{ type String, required true, trim true, minlength [2, Full name must be at least 2 characters], maxlength [50, Full
name cannot exceed 50 characters], }, password { type String, required [true, Password is required], minlength
[6, Password must be at least 6 characters], }, }, { timestamps true } ); JWT Authentication Methods import jwt
from jsonwebtoken; export function generateJWT(user) { const payload = { id user.id, username user.username,
email user.email }; return jwt.sign(payload, processenvJWT_SECRET, { expiresIn 1d }); } 32 export function
authenticateJWT(req, res, next) { const token = req.headers.authorization?.split( )[1]; if (!token) return
res.status(401).json({ error Access token missing }); jwt.verify(token, processenvJWT_SECRET, (err, user) => { if
(err) return res.status(403).json({ error Invalid token }); requester = user; next(); }); } FHE Integration and Security
Implementation Guide 623 FHE Integration Implementation FHE Service Class class FHEService {
constructor() { this.initialized = false; this.module = null; this.simulationMode = false; } async initialize() { try {
console.log( Initializing FHE Service for MCP HTTP Wrapper); const fheDir = path.join(__dirname, , fhe); const
jsFile = path.join(fheDir, openfhe_pke_es6js); const wasmFile = path.join(fheDir, openfhe_pke_es6wasm); if
(!fs.existsSync(jsFile) || !fs.existsSync(wasmFile)) { console.warn( FHE WebAssembly files not found, falling
back to simulation mode); this.simulationMode = true; this.initialized = true; 33 return true; } // Dynamic import
of FHE module const FHEModule = await import(jsFile); this.module = await FHEModule.default(); console.log(
FHE Service initialized successfully); this.initialized = true; return true; } catch (error) { console.error( FHE
initialization failed:, error.message); this.simulationMode = true; this.initialized = true; return false; } } async
encryptData(data) { if (!this.initialized) { await this.initialize(); } if (this.simulationMode) { return { encrypted
true, data `sim_encrypted_${Buffer.from(data).toString(base64)}`, algorithm simulation, timestamp new
Date().toISOString() }; } try { 34 // Actual FHE encryption implementation const encrypted =

```

thismoduleencrypt(data); return { encrypted true, data encrypted, algorithm BGV-FHE, timestamp new Date().toISOString() }; } catch (error) { console.error(Encryption failed:, error); throw new Error(Failed to encrypt data); } } } Explanation The FHEService class implements a Fully Homomorphic Encryption service with intelligent fallback capabilities.

Heres how it works Constructor & Initialization □ The service starts uninitialized and checks for required WebAssembly files (OpenFHE PKE ES6 modules) □ If the required js and wasm files are missing, it gracefully falls back to simulation mode □ This ensures the application continues to function even in development environments without full FHE setup Dynamic Module Loading □ Uses dynamic imports to load the FHE WebAssembly module only when needed □ This approach prevents blocking the main application startup and allows for conditional loading Encryption Process □ In simulation mode Creates a base64-encoded placeholder that mimics encrypted data structure □ In full FHE mode Uses the BGV (Brakerski-Gentry-Vaikuntanis) scheme for actual homomorphic encryption 35 □ Both modes return consistent data structures with metadata including algorithm type and timestamp Error Handling □ Comprehensive try-catch blocks ensure service remains available even if individual operations fail □ Automatic fallback to simulation mode prevents complete service failure □ Detailed logging helps with debugging and monitoring 624 Security Middleware Implementation Security Headers and Rate Limiting import helmet from helmet; import rateLimit from express-rate-limit; import slowDown from express-slow-down; import mongoSanitize from express-mongo-sanitize; import compression from compression; // Security headers middleware export const securityHeaders = (req, res, next) => { res.setHeader(X-Content-Type-Options, nosniff); res.setHeader(X-Frame-Options, DENY); res.setHeader(X-XSS-Protection, 1; mode=block); res.setHeader(Referrer-Policy, strict-origin-when-cross-origin); res.setHeader(Permissions-Policy, geolocation=(), microphone=(), camera=()); next(); }; // Rate limiting configuration export const rateLimiters = { general rateLimit({ windowMs 15 * 60 * 1000, // 15 minutes max 1000, // Limit each IP to 1000 requests per windowMs message {36 error Too many requests from this IP, please try again later, retryAfter 15 * 60 // 15 minutes in seconds }, standardHeaders true, legacyHeaders false, }), auth rateLimit({ windowMs 15 * 60 * 1000, // 15 minutes max 10, // Limit each IP to 10 auth requests per windowMs message { error Too many authentication attempts, please try again later, retryAfter 15 * 60 }, skipSuccessfulRequests true }) }; // Security middleware collection export const securityMiddleware = { helmet helmet({ contentSecurityPolicy { directives { defaultSrc [self], styleSrc [self, unsafe-inline], scriptSrc [self], imgSrc [self, data:, https:], connectSrc [self, https://apiopenai.com, https://apianthropic.com], fontSrc [self], objectSrc [none], mediaSrc [self], frameSrc [none], }, }, 37 crossOriginEmbedderPolicy false }), mongoSanitize mongoSanitize(), compression compression() }; Explanation This security middleware implementation provides comprehensive protection for the HTTP wrapper through multiple layers of defense Security Headers Middleware □ X-Content-Type-Options nosniff Prevents browsers from MIME-type sniffing, reducing XSS attack vectors □ X-Frame-Options DENY Blocks the page from being embedded in frames, preventing clickjacking attacks □ X-XSS-Protection Enables browsers built-in XSS filtering (legacy support) □ Referrer-Policy Controls how much referrer information is shared with external sites □ Permissions-Policy Restricts access to sensitive browser APIs like geolocation and camera Rate Limiting Strategy □ General Rate Limiter Allows 1000 requests per 15-minute window per IP address, suitable for normal API usage □ Authentication Rate Limiter Stricter limit of 10 auth attempts per 15 minutes to prevent brute force attacks □ skipSuccessfulRequests true for auth limiter means successful logins dont count against the limit □ Standard headers provide clients with rate limit information for better handling Comprehensive Security Middleware □ Helmet Configuration Advanced Content Security Policy (CSP) that allows connections only to trusted AI API endpoints (OpenAI, Anthropic) □ MongoDB Sanitization Automatically removes or escapes MongoDB operators from user input to prevent NoSQL injection attacks □ Compression Reduces response size for better performance while maintaining security □ CSP Directives Strictly controls resource loading - only self-hosted resources allowed except for specific HTTPS images and trusted API connections This multi-layered approach ensures the application is protected against common web vulnerabilities while maintaining functionality for legitimate users.

38 625 Database Connection and Configuration MongoDB Connection with Error Handling import mongoose from mongoose; import LoggingService from /services/loggingServicejs; const connectDB = async () => { try { const MONGODB_URI = process.env.MONGODB_URI || mongodb://localhost:27017/secure-bridge; LoggingServiceinfo(Attempting to connect to MongoDB, { uri MONGODB_URIreplace(/\\/*/@/, /****:***@)

```
// Hide credentials in logs }); const connectionOptions = { useNewUrlParser true, useUnifiedTopology true,
maxPoolSize 10, serverSelectionTimeoutMS 5000, socketTimeoutMS 45000, bufferCommands false,
bufferMaxEntries 0 }; const connection = await mongooseconnect(MONGODB_URI, connectionOptions);
LoggingServiceinfo( MongoDB connected successfully, { host connectionconnectionhost, port
connectionconnectionport, database connectionconnectionname }); 39 // Handle connection events
mongooseconnectionon(error, (error) => { LoggingServiceerror(MongoDB connection error:, error); });
mongooseconnectionon(disconnected, () => { LoggingServicewarn( MongoDB disconnected); }); return
connection; } catch (error) { LoggingServiceerror( MongoDB connection failed:, { error errormessage, stack
errorstack }); // Dont exit process in development if (processenvNODE_ENV === production) { processexit(1);
} throw error; } }; export default connectDB; Key Features FHE Service □ Graceful Fallback Automatically
switches to simulation mode if WebAssembly files are unavailable □ Dynamic Module Loading Loads FHE
modules on-demand to optimize performance □ Error Handling Comprehensive error handling with detailed
logging □ BGV Algorithm Uses the BGV (Brakerski-Gentry-Vaikuntanis) FHE scheme 40 Security Middleware
□ Rate Limiting Different limits for general requests (1000/15min) and authentication (10/15min) □ Security
Headers Comprehensive set of security headers including CSP, XSS protection □ Input Sanitization MongoDB
injection protection and data sanitization □ Compression Built-in response compression for performance
Database Configuration □ Connection Pooling Optimized connection pool settings for production use □
Timeout Handling Appropriate timeouts for server selection and socket operations □ Environment Awareness
Different behavior for development vs production environments □ Event Monitoring Real-time monitoring of
connection status and errors This implementation provides a robust foundation for secure, encrypted data
handling with proper fallback mechanisms and comprehensive security measures.
```

41 63 Screenshot Home page Figure 5 home page Login page Figure 6 login page 42 Signup page Chat page
Figure 7 Signup page Figure 8 chat page 39 43 API key management page Admin login page Figure 9 Api key
management Figure 10 admin login 44 Admin Dashboard page Admin panel user management Figure 11 admin
Dashboard page Figure 12 Admin panel user management 45 Admin system health monitor page Admin System
Settings page Figure 14 admin system setting Figure 13 Admin system health monitor page46 Chapter-7
SOFTWARE TESTING 71 Unit Testing Unit testing verifies individual components (functions, classes,
endpoints) in isolation so each unit meets its specification and defects are caught early.

Tests run in a controlled environment using mocks/stubs for databases, external APIs, and crypto engines so
failures point to the unit not dependencies.

Deterministic, repeatable runs with ephemeral data and no live network make suites fast and reliable, ideal for
continuous integration.

Coverage thresholds (statements/branches/functions/lines) act as quality gates, with higher targets on critical
paths like auth, encryption, and access control.

For security/crypto-heavy systems such as Secure Bridge, simulation modes validate encryption/FHE logic, and
negative tests harden inputs, headers, and rate limits.

Table 71 Unit Testing Module TC ID Scenario Expected Result Status Notes Authentication AUTH- 001 User
registration with valid inputs Account created; verification/JWT issued Pass Happy path AUTH- 002 Login with
valid credentials Auth success; access token; dashboard Pass Happy path AUTH- 003 Invalid login attempts
Clear error; no token; lockout after threshold Pass Security AUTH- 004 JWT structure & expiry Valid token
resolves user context; expired is rejected Pass Security 47 Module TC ID Scenario Expected Result Status Notes
Encryption (FHE) FHE-001 Encrypt message Ciphertext envelope produced per policy Pass Crypto correctness
FHE-002 Decrypt valid ciphertext Original plaintext restored Pass Crypto correctness FHE-003 Simulation
fallback Simulation mode activates when WASM unavailable Pass Resilience FHE-004 Malformed envelope
Error raised; no processing Pass Input hardening API Keys API-001 Key creation & storage Key persisted
securely; metadata saved Pass Secrets hygiene API-002 Key format validation Invalid formats rejected with
message Pass Validation API-003 Key rotation New key active; old revoked gracefully Pass Operational safety
API-004 Usage rate limiting Excess usage throttled (429 + Retry-After) Pass Abuse control MCP Service MCP-
001 Tool discovery Complete, typed tool list returned Pass Contract MCP- 002 Weather tool exec Structured data
returned for city query Pass Functionality MCP- 003 Encrypted param exec Tool operates without server
decryption Pass Privacy by design MCP- 004 Error handling Invalid params → descriptive 4xx error Pass
Robustness Database DB-001 User save Document persisted; id returned Pass CRUD DB-002 Query sanitization

Injection payloads neutralized/blocked Pass Security 48 Module TC ID Scenario Expected Result Status Notes
DB-003 Index & perf checks Query completes within target SLA Pass Performance DB-004 Connection
resilience Auto-reconnect on transient failures Pass Reliability Security MW SEC-001 Input sanitization (XSS)
Payload neutralized; safe rendering Pass Security SEC-002 CORS policy Disallowed origins blocked Pass
Security SEC-003 IP-based rate limits Excess requests limited appropriately Pass Security SEC-004 Security
headers CSP, Referrer-Policy, X-Frame-Options set Pass Compliance Chat CHAT- 001 Message processing
pipeline AI response rendered to user Pass Core flow CHAT- 002 History persistence Encrypted content stored;
metadata visible Pass Data lifecycle CHAT- 003 Multi-provider switch Valid responses across providers Pass
Interop CHAT- 004 FHE in chat Ciphertext only on server; client decrypts Pass Privacy Admin ADMIN- 001
User management Create/modify/delete reflected immediately Pass RBAC covered ADMIN- 002 System
metrics Health status and metrics accurate Pass Observability ADMIN- 003 Settings management Config
saved/applied; audit recorded Pass Governance ADMIN- 004 Audit logs Privileged actions logged with
actor/IP/time Pass Compliance 49 Module TC ID Scenario Expected Result Status Notes Performance PERF-
001 API response time p95 within ≤ 500 ms target Pass SLO PERF- 002 Concurrent users ≥ 100 users without
degradation Pass Scale PERF- 003 Soak memory No leaks; stable heap over run Pass Stability PERF- 004 DB
heavy queries Aggregations within bounds; indexes used Pass SLO 72 Automation Testing Automation
orchestrates linting, unit, integration, E2E smoke, and security scans in a CI pipeline so every change is validated
quickly and consistently.

Ephemeral stacks provide production-like isolation without flakiness.

Integration suites exercise real HTTP to the local stack while chaos injectors verify resilience.

Performance jobs watch p95/p99 latency and resource usage under load.

Security jobs enforce headers, token rules, and input hardening.

A coverage gate blocks merges that weaken critical paths.

Table 72 Status Area Automation Scope (short) Pass Criteria Status CI Pipeline Lint $\rightarrow$ Unit $\rightarrow$ API/Contract $\rightarrow$
E2E smoke $\rightarrow$ Sec scans $\rightarrow$ Coverage All green; coverage $\geq$ thresholds; build fails on critical vulns Passing
Integration Real HTTP to local stack; DB container; external mocks + chaos; FHE sim; WASM init error
Contracts valid; DB ops correct; retries/fallbacks OK; FHE round-trip OK Passing Performance 100+ concurrent
R/W; heap snapshots; p95/p99 for chat/MCP/admin; step & burst p95 within SLO; errors $\leq$ target; no leaks;
graceful 429 Passing 50 Area Automation Scope (short) Pass Criteria Status Security Vuln scans; header/CSP
audit; injection/XSS/path traversal; auth tokens; rate limits No criticals; headers correct; attacks blocked; invalid
tokens rejected Passing End-to-End Register $\rightarrow$ verify $\rightarrow$ login $\rightarrow$ encrypted chat $\rightarrow$ history; MCP flow; Admin ops;
cross-browser WASM Flows succeed; ciphertext stored; RBAC enforced; graceful degradation Passing 73 Test
Cases A) Test Cases Table 73 TC ID Module Scenario Expected Result Status UT- 001 Encryption
Encrypt $\rightarrow$ Decrypt round-trip Plaintext recovered; auth tag valid Pass UT- 002 Encryption Reject non-approved
params Error code; no ciphertext emitted Pass UT- 003 Envelope Validate schema fields Invalid/missing fields
 $\rightarrow$ 4xx error Pass UT- 004 Security URL allow/block Only allowlisted schemes/domains pass Pass UT- 005
Security IP blocking RFC1918/localhost/link-local denied Pass UT- 006 DB Security Operator sanitization
\$where/\$gt/\$ne stripped/blocked Pass UT- 007 Headers CSP/Referrer-Policy/XFO set Required headers present
& correct Pass UT- 008 User Model Registration & hashing Constraints enforced; hash not reversible Pass 51 TC
ID Module Scenario Expected Result Status UT- 009 API Key Model Create/rotate/deactivate New key active;
old gracefully revoked Pass UT- 010 Settings Model Rate-limit & flags Values persisted; enforced at runtime
Pass UT- 011 Chat Session TTL & backup metadata Expired docs purged; backup fields valid Pass UT- 012
MCP Routing Method & schema validate Bad schema $\rightarrow$ mapped structured error Pass B) Integration Test Cases
Table 74 TC ID Flow / Endpoint(s) Scenario Expected Result Status IT- 001 /register $\rightarrow$ /login Auth happy path
JWT issued; redirect dashboard Pass IT- 002 /login (lockout) Excess invalid attempts Account temp locked; audit
entry Pass IT- 003 /chat/completions FHE on encrypt $\rightarrow$ compute $\rightarrow$ decrypt Server handles ciphertext; client
decrypts Pass IT- 004 /chat/history Persist & retrieve Encrypted content; metadata visible Pass IT- 005 Provider
toggle OpenAI $\leftrightarrow$ Anthropic $\leftrightarrow$ Google Valid responses across providers Pass IT- 006 /admin/* RBAC matrix
Roles restrict actions & views Pass IT- 007 /api/weather MCP tool execute Structured payload; rate-limited Pass
IT- 008 Backup/Restore Chat DB restore test Data recovered; indexes intact Pass 52 C) Security / Negative Test
Cases Table 75 TC ID Vector Scenario Expected Result Status SEC-001 Auth JWT tamper / invalid token
Rejected with 401; no data leak Pass SEC-002 Input Hardening NoSQL payloads Sanitized/blocked; safe queries

Pass SEC-003 XSS Script/HTML injection Neutralized; no DOM execution Pass SEC-004 SSRF Malicious URL targets Blocked by URL/IP rules Pass SEC-005 Path Traversal / in paths Denied; normalized safe path Pass SEC-006 Rate Limiting Rapid bursts 429 with Retry-After header Pass SEC-007 Session Mgmt Idle & absolute timeouts Session expires as configured Pass SEC-008 Headers CSP/referrer/XFO audit Present; strict values Pass D) Performance Test Cases Table 76 Performance TC ID Target Load Pattern Success Metric (SLO) Expected / Result PERF- 001 Chat API p95 Steady 50 RPS p95 $\leq$ 500 ms; errors $\leq$ 1% Pass PERF- 002 MCP p95 Steady 30 RPS p95 $\leq$ 600 ms Pass PERF- 003 Concurrent users 100+ virtual users No degradation; stable throughput Pass PERF- 004 Burst resilience 5 $\times$ spike for 60 s Graceful 429 + Retry-After Pass PERF- 005 Memory 60-min soak No leaks; heap stable Pass PERF- 006 Chat DB queries History/aggregations p95 within target; indexes used Pass 53 Chapter-8 CONCLUSION Secure Bridge demonstrates that a privacy-first AI chat platform capable of rich tool use can be engineered without exposing user plaintext to servers at any point in the workflow.

By combining client-side encrypt-before-send, a homomorphic computation path for supported operations, and a confidential-compute fallback for complex or latency-sensitive tasks, the system delivers practical functionality while upholding strong confidentiality guarantees.

The Model Context Protocol (MCP) gateway acts as the policy and safety boundary for tool integrations, enforcing typed I/O, capability scoping, redaction, rate limits, and auditable decisioning.

Together, these choices transform privacy from a promise into a verifiable property of the system.

From the user's perspective, Secure Bridge behaves like a modern chat application messages stream, conversations are organized, and tools can be invoked to fetch weather, query knowledge, or perform analysis. Under the surface, message content is always ciphertext at the platform boundary, and only minimal, non-sensitive metadata is retained to support usability (titles, timestamps, optional tags).

History retrieval, export, and deletion respect the same zero-knowledge constraints; backup and restore procedures operate on encrypted artifacts; and disaster-recovery drills validate that confidentiality is preserved even under failure modes.

The architecture makes explicit trade-offs to keep privacy and operability in balance.

The FHE path supports arithmetic circuits and selected transformations with acceptable p95 latencies; where operations exceed those bounds or require non-homomorphic primitives, requests are routed to a trusted execution environment (eg, SGX/TDX) with attestation and strict data-minimization.

The MCP gateway ensures that every tool invocation passes through schema validation, least-privilege credentials, timeouts, retries with backoff, and circuit breakers, so that failures degrade safely and predictably. On the operations side, the admin panel converts security posture into concrete controls user lifecycle, rate limiting, maintenance windows, backup/restore, and encryption profile selection without ever revealing user plaintext to operators.

Security is layered and measurable.

Transport and at-rest encryption are non-negotiable; keys for infrastructure secrets live in managed KMS, while user FHE private keys never leave the client.

Authentication and authorization follow role-based access control with scoped tokens; privileged actions require elevated roles and are always recorded in append-only audit logs with actor, time, and redacted parameters.

Policy changes, incident responses, and backup verifications produce 54 traceable artifacts, enabling routine compliance checks and external audits.

Observability is engineered to be privacy-preserving structured logs contain correlation IDs and metrics but never message content, and dashboards surface health, latency, and error rates without compromising confidentiality.

Engineering quality is not an afterthought but a first-class requirement.

The codebase is modular, containerized, and delivered via CI/CD with automated linting, unit tests, API/contract tests, E2E journeys, and security scans.

Simulation mode allows deterministic testing of encryption workflows where WASM or enclaves would otherwise complicate repeatability.

Performance baselines and SLOs such as service uptime, p95/p99 latency envelopes for homomorphic operations, and error-budget policies are defined up front and enforced by alrts.

These practices make the system evolvable features can be added behind flags, rolled out gradually, and rolled

back safely if regressions appear.

The data model is deliberately conservative.

Only ciphertext is stored for message content; indices are built on metadata designed for discoverability without content exposure.

TTL policies govern retention; archival and legal hold can be satisfied using encrypted snapshots; and export flows deliver user-controlled packages that remain encrypted end-to-end.

This restraint has UX implications semantic search across plaintext is intentionally unavailable but it is the principled cost of keeping user privacy non-negotiable.

Practical limits and risks are acknowledged.

Today's FHE remains computationally intensive, and interactive latencies depend on careful circuit design and, where available, hardware acceleration (GPU/FPGA).

Trusted execution environments introduce a different trust anchor that must be maintained microcode updates, attestation verification, and enclave configuration hardening.

Tool ecosystems evolve quickly; policies, schemas, and scopes require ongoing governance to prevent privilege creep or data over-collection.

Finally, multi-tenant scale magnifies the importance of isolation, quota management, and fair-use controls to protect shared resources.

These constraints define a clear roadmap for improvement.

Hardware-assisted FHE and bootstrapping-friendly schemes can expand the set of real-time workloads that never leave ciphertext form.

Richer policy automation in MCP context-aware redaction, risk-based prompts, and trust-tiered tool catalogs can reduce operator burden while tightening guardrails.

Enterprise capabilities such as SSO, SCIM provisioning, per-tenant key hierarchies, and fine-grained data residency will broaden adoption.

On the client side, mobile apps and browser extensions can extend encrypted capture and display into more user contexts, while keeping keys and plaintext strictly local.

Formal compliance programs (SOC 2, HIPAA, ISO 27001) can build on the existing auditability, key management, and data-minimization foundations already present.

Most importantly, Secure Bridge shows that privacy and capability are not mutually exclusive.

With a zero-knowledge data model, a policy-centric integration layer, and a dual compute path that privileges confidentiality by default, the platform delivers useful AI-assisted interactions without compromising user trust. The combination of verifiable controls, observable behavior, and repeatable tests turns privacy from a marketing claim into an operational fact.

As acceleration technologies mature and governance patterns deepen, the platform is positioned to serve broader workloads at higher performance while holding fast to the principle that user data belongs to users, even while it is being processed.

CHAPTER-9 Future Enhancements

This section lays out a simple, implement-first roadmap for Secure Bridge.

The items are grouped so a student team can deliver them in short sprints.

Each paragraph explains what to add, how to build it at a basic level, and how we will know it worked without requiring extra tooling or complex infrastructure.

91 Privacy and Cryptography

The near-term privacy goal is to keep plaintext off the server while still offering a usable chat experience.

The first step is to ship client-side encryption with a visible "FHE mode" toggle.

Keep the default in simulation mode for development; when FHE is available in the browser, enable the real path.

Provide two parameter presets ("safe" and "fast") and handle failures by falling back to the simulation path without crashing the UI.

Success is measured by being able to send/receive messages in both modes with no server logs containing plaintext.

Next, expand supported operations carefully.

Begin with homomorphic add/multiply and batching (enough for simple scoring or filters), then explore a small CKKS-style simulated vector operation (such as a dot-product for relevance) to prove the path for encrypted

similarity.

The aim is not to perfect accuracy, but to keep latency acceptable and avoid any server-side decryption.

We will consider this complete when a demo page runs a small vector computation on ciphertext and returns a useful score.

Finally, protect data beyond the live system.

Enable ciphertext-only backups with a restore script, and set retention policies (TTL) per collection so expired chats are removed automatically.

Quarterly test a backup/restore round trip if the restored collection still has only ciphertext in message bodies, the control works.

92 Secure Compute and Execution Some tasks will not fit the FHE path in a typical student environment.

To handle these safely, add a “fallback worker” a small, separate service that performs non-FHE operations and returns only what the UI needs.

Route requests to this worker when depth or size would make the FHE path too slow.

Document the policy that decides routing (for example “if estimated cost > threshold, use fallback”).

We know this is working when large prompts return results via the fallback without failing the main API.

57 Alongside this, sandbox tool execution.

Run external tools in a container or WASM sandbox with a tight network allowlist, limited filesystem, and timeouts.

Log and block any egress the policy does not permit.

The enhancement is done when an intentionally misconfigured tool cannot reach a blocked domain and the attempt is visible in the audit view.

93 Data Governance and the Chat Database Bring the data model closer to production practices without over-engineering.

Enforce ciphertext- only storage for message bodies through schema validation; allow only minimal metadata (timestamps, tags) for queries and indexing.

Add per-tenant data segregation in the schema (tenant ID on all rows) to prepare for multi-user or multi-class demos.

Add retention controls (TTL indexes) and a legal-hold flag that temporarily disables deletion for selected records.

We will validate this by scanning stored documents for plaintext fields (there should be none), verifying index-based searches return expected sets, and checking that legal-hold prevents deletion until the flag is lifted.

As an optional stretch, introduce a basic per-tenant key model with rotation exposed in the admin UI.

Complete the feature when rotating a tenant key does not interrupt active sessions and new messages are written under the new key.

94 Product and User Experience Focus on making the system easy to try.

Build a responsive chat UI that works on phones, add “export chat (encrypted)” so users can back up their own data, and provide readable error messages when FHE is unavailable or a tool is rate-limited.

Convert the site into a Progressive Web App so it can install on mobile and cache the shell for quick loads; only non-sensitive assets should be cached.

A lightweight browser extension adds value without large scope.

It should capture selected text and the current URL, encrypt on the client, and send through the same gateway with Model Context Protocol (MCP) rules enforced.

The feature is done when one click from the extension sends an encrypted message and the admin audit shows a redacted, policy-compliant record of the action.

95 Enterprise and Compliance Foundations For coursework and demos, aim for simple and clear controls rather than a full compliance stack.

Implement role-based access control in the admin panel (Admin vs.

Super Admin) and protect 58 all administrative routes.

Add append-only audit logging for privileged actions (who did what, when, and from where), ensuring no message content is included.

These basics support future alignment with formal programs and make the system easier to grade and maintain.

We consider this complete when a non-admin cannot access admin pages and when actions like “change

settings” or “rotate key” produce audit entries.

96 Observability and Reliability Operate the system safely without exposing content.

Add a /health endpoint that reports liveness, simple dependency checks, and version.

Track a few privacy-preserving metrics request count, error count, and average/percentile latency by endpoint.

On tool calls, implement bounded retries with jitter and a circuit breaker to prevent cascading failures.

Display health and metrics on the admin dashboard with green/red indicators and simple charts.

Define two small SLOs for the student deployment (1) p95 chat round-trip time under a modest load (for example, ≤ 700 ms in simulation mode with short prompts), and (2) error rate under load (for example, $\leq 3\%$ at 100 concurrent users).

Use these numbers only to guide tuning; the goal is to spot regressions early rather than to guarantee production-grade uptime.

97 MCP and Tooling Ecosystem Treat tools as first-class, governed integrations.

For each tool, publish a typed schema for inputs and outputs, enforce least-privilege scopes, and rate-limit per tool and per user.

Keep a small catalog (for example, Weather plus one LLM provider) that a student can understand quickly.

Add a “policy as code” file to the repository (a JSON or YAML policy) and require a basic test for any change.

The enhancement is done when invalid inputs are rejected with a clear 4xx response, over-limit requests return 429, and tool changes must pass a simple policy test in CI before merging.

98 Cost and Performance Even on a classroom budget, performance can improve with basic steps.

Cache non-sensitive metadata for list views (recent chats, users) to keep admin pages fast.

On the client, enable WASM SIMD and web workers where supported to reduce perceived latency for cryptographic operations.

In the cluster or VM, configure horizontal scaling for the API and MCP workers so the system can add a replica under load.

Declare success when admin list views consistently render under a few hundred milliseconds, client-side computation time drops for supported browsers, and scaling adds capacity without service restarts.

59 99 Research and Community Finally, contribute a small, repeatable benchmark that reflects chat-like workloads (short ciphertexts, batching, a simple homomorphic transform).

Publish the script and a one-page readme so others can reproduce results and compare settings.

Use the benchmark to guide parameter presets and to justify changes to batching or routing policies.

When the script produces a stable chart and helps catch a regression (for example, a parameter set that increases latency), this effort has paid off.

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